

ANIL PRADHAN

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AI & Backend Engineer specializing in autonomous multi-agent architectures, computer vision pipelines, and resilient backend systems with Python, FastAPI, LangGraph, OpenCV, and YOLO11n.

Dell FutureMinds National Grand Finalist (Top 6 / 8,000+) · 4× AI Hackathon Builder (Dell, MSFT, AWS, Google)

EDUCATION

Institute of Technical Education & Research (ITER), SOA University

Bachelor of Technology (B.Tech), Computer Science & Engineering

2024 – 2026

CGPA: 8.93

Nalanda Vidya Mandir Higher Secondary School, Berhampur, Odisha

Class XII (Higher Secondary) — Physics, Chemistry, Mathematics with IT

2022 – 2024

Score: 85%

Odisha Adarsha Vidyalaya (OAV), Digapahandi, Ganjam, Odisha

Class X (CBSE Board)

2020 – 2022

Score: 78.8%

TECHNICAL PROJECTS

VisionForge AI

[GitHub](#) — AI/ML & Pipeline Engineer | Team: Disha & Anil

Aug 2026 – Present

- Built an autonomous inspection pipeline using **LangGraph state machines** to run OCR, seal-matching, and YOLO11n detection in parallel, with **Groq and Gemini Vision** models giving the final pass/fail verdict in under 5s.
- Trained and fine-tuned a custom **YOLO11n object-detection model on 4,448 images**, reaching **98.4% mAP@50** for identifying and locating hardware parts down to 12×12px.
- Designed a custom evidence-fusion scoring method (**Anomaly Max-Pooling**) combined with EasyOCR text verification and Gemini embeddings + FAISS similarity search for reference matching — so no defect gets diluted or missed — validated with **203 automated tests**.
- Architected **sub-30ms defensive fast-fail gates** using Laplacian blur variance and Error Level Analysis (ELA) to eliminate wasted compute on blurred or manipulated uploads prior to neural inference.

Tech: Python · FastAPI · LangGraph · YOLO11n · OpenCV · FAISS · Groq LPU · Gemini 3.5 Flash

TaskPilot AI

[Live Demo](#) | [GitHub](#) — Integration Lead | Team IdeaForg-E

Jul 2026 – Present

- Built a **6-stage multi-agent pipeline** (Agent 0 Orchestrator) in Python & FastAPI to collect, extract, clean, and schedule tasks from 5 tools in **~20 seconds**.
- Developed **smart deduplication and a 7-factor priority engine** (severity, deadline, outage risk, customer impact) with a 7-point quality checker to rank tasks with clear explanations.
- Implemented a **reliable LLM client with circuit breaker** (Groq + local fallback rules) and an AI chat assistant for **instant P1 bug injection** and calendar-aware daily scheduling.

Tech: Python · FastAPI · Multi-Agent Systems · SQLAlchemy · SQLite (WAL) · Groq · React 18

AI Career Mentor

[Live Demo](#) | [GitHub](#) — Solo Developer

Feb 2026 – Aug 2026

- Built a multi-agent AI system using **LangGraph** to orchestrate 4 parallel agents (Resume, Market, Roadmap, LinkedIn) with shared state, reducing analysis latency by **~60%**.
- Developed a real-time mock interview engine with a **7-phase FSM over WebSockets**, featuring a live code editor and resume-personalized questions.
- Designed a fault-tolerant backend (**FastAPI, PostgreSQL, Redis**) with **RAG-powered roadmap generation via ChromaDB**, multi-provider LLM failover, and a **113-test CI/CD pipeline**.

Tech: FastAPI · LangGraph · Next.js · ChromaDB · Redis · Docker · ONNX

TRAINING

Agentic AI Training Program — CogniForce.AI & ITER, SOA University

2026 – Present

- Hands-on training in **Agentic AI, LLMs, prompt engineering, RAG, tool calling, MCP**, and multi-agent orchestration.
- Built and experimented with autonomous agents using **OpenClaw, Multica, AnythingLLM, OpenRouter, Claude Code**, and custom MCP servers.

TECHNICAL SKILLS

Languages: Python, Java, SQL

AI & Computer Vision: LangGraph, Multi-Agent Systems, RAG, Prompt Engineering, LLM Integration, OpenCV, YOLO11n, OCR, Anomaly Detection (SSIM)

Backend & Databases: FastAPI, REST APIs, SQLAlchemy, JWT/RBAC, PostgreSQL, Redis, ChromaDB, SQLite

Core CS Concepts: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Developer Tools: Git/GitHub, Docker, Linux Commands, Postman

ACHIEVEMENTS & CERTIFICATIONS

- Dell FutureMinds AI Hackathon 2026 — National Grand Finalist & Certificate of Excellence:** Selected among Top 6 Grand Finalist teams nationwide out of 8,000+ participants across 4 competitive stages (Screening → Top 370 Hackathon → Top 10 → Top 6). Flew to Dell Bengaluru Headquarters to present our autonomous hardware inspection solution live before Dell's senior engineering leadership and jury.
- FlowFinance × Techfest IIT Bombay — Round 2 Qualifier:** Designed a GTM strategy deck (platform analysis, customer personas, acquisition/retention framework) for FlowFinance (AI SMB fintech platform) with real customer outreach.
- 4× AI Hackathon Builder:** Built and deployed AI solutions across 4 hackathons, including Microsoft AI Dev Days, Amazon Nova Challenge, and Google Solution Challenge.
- 100+ Day DSA Streak:** Maintained a 100+ day coding streak on GeeksforGeeks' 160 Days of Code Challenge, solving DSA problems daily — recognized with GFG consistency reward.
- Certifications:** IBM AI Fundamentals · Cisco Python Essentials 2 · Wells Fargo SWE (Forage) · HackerRank (Problem Solving, SQL, Java, Python).